DPFF-Net: Dual-Polarization Image Feature Fusion Network for SAR Ship Detection

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***Abstract*— Intelligent ship detection algorithms for synthetic aperture radar (SAR) images have achieved significant results in Earth observation applications. By learning features such as scale, shape, and texture from samples, they can quickly locate and recognize ships in complex backgrounds. However, due to the lack of use of polarization features, the upper bound of detection performance is still limited, especially under poor image quality conditions such as ambiguous interference. To solve this, the dual-polarization image feature fusion network (DPFF-Net) is proposed. The key of it lies in adaptive mining, enhancement, and fusion of polarization features through the designed siamese structure, polarization-aware feature enhancement block (PAEB), and dynamic gated fusion block (DGFB). With fully utilizing complementary information hidden between copolarization and cross-polarization data, more comprehensive and accurate features are obtained and used as the detect head input. Thus, the proposed algorithm achieves state-of-the-art performance, and its effectiveness is validated by experiments on dual-polarization SAR datasets.**

***Index Terms*— Dual-polarization, feature fusion, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR), which holds the specific advantages of all-day and all-weather work, is extensively applied in various Earth observation tasks. With rapid development of spaceborne SAR imaging technologies, SAR data has steadily expanded in quantity and quality for the past few years, which contains considerable research significance and practical values to conduct ship detection in high-resolution wide-swath SAR images.

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Traditional detection methods for SAR ship mainly locate and extract the concerned targets in stages. These stages include preprocessing, region screening, and target discrimination [1]. Among them, the region screening stage is more critical. In this stage, a series of algorithms [2], [3], [4], [5] based on constant false alarm rate (CFAR) emerged. The CFAR detection methods mainly invert the background distribution information of the sea surface through parameter estimation based on classical statistical models, and screen out targets different from the background distribution through set thresholds. These algorithms work well for simple ocean scenes containing only ships but fail for complex near-shore scenes where the background distribution is challenging to model statistically. In addition, joint optimization in stages is complicated, further limiting traditional ship detection capabilities.

Deep learning has recently flourished in the field of SAR ship detection due to its ability to mine deep features of objects adaptively, represent objects robustly and detect objects end-to-end. Aiming at the multiscale characteristics of ship targets, Dai et al. [6] construct a fusion feature extractor network based on Faster region-CNN (R-CNN) to fuse features in a bottom-up and top-down manner. Multiscale ships' location and semantic features are enhanced, especially for small SAR ships. Due to the horizontal bounding box (HBB) representation containing much background interference limits the performance of subsequent tasks, Zhang et al. [7] propose an oriented normalized Gaussian function in the regression subnetwork to characterize the center point of targets and regress bounding boxes, effectively improving the recall rate of the model. Considering that the imaging conditions easily affect the characteristics of SAR targets, Fu et al. [8] propose a scattering key points guided feature adaption method, which can increase the localization accuracy and the model adaptability. Based on the SAR amplitude data of a single polarization, the above methods continuously establish a more comprehensive representation model from the detailed semantic information fusion and the stable scattering position information, improving the target interpretation accuracy. However, due to the lack of mining and expression of target polarization features, the above methods still have insufficient detection ability when applied in complex scenes, and some false targets are challenging to be filtered out, such as the "ghosts" generated by azimuth ambiguities.

Some studies indicate that the SAR ship detection is a highly complex task [9], [10], [11]. The information contained in

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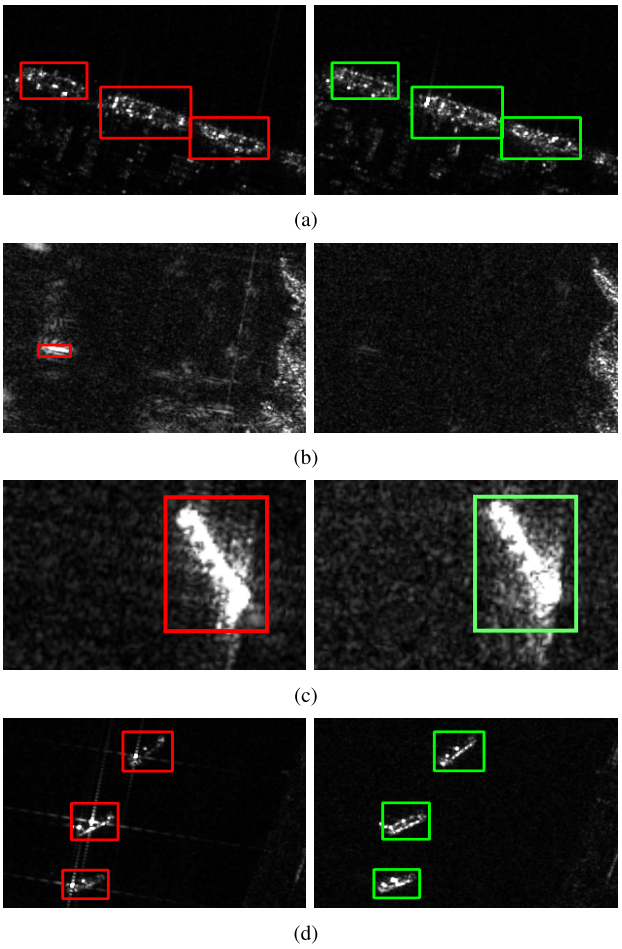

Fig. 1. Comparison of characteristics under different polarization modes. The left and right columns present the copolarization and cross-polarization images, respectively. (a) Difference of scattering intensity. (b) Azimuth ambiguity. (c) and (d) Sidelobe characters of clustered and isolated strong scattering points.

one single polarization channel (HH, HV, or VV) could not fully characterize the scattering mechanisms of the target [11]. It is not generally adequate to effectively monitor metallic targets [10], which leads to the difficulty of optimizing the detection algorithms using only one polarization mode [9]. Under different polarization modes, imaging parameters such as generalized-K (GK) distribution have various sensitivities to ocean scenes containing targets [12], and the backscattering of ship targets varies with incidence angles [13], which is what we need to consider and exploit. Some researches note that the backscatter from the ocean for HV polarization should be distinct from HH or VV polarization, and consequently, their ship–sea contrast should be different. HV provides the better ship–sea contrast and ship detectability than HH at low incidence angles, and HH is better at large incidence angles [14], [15], [16], [17]. As shown in Fig. 1(a), the ship–sea contrast looks higher in cross-polarization mode. Additionally, due to the azimuth ambiguity caused by the limited sampling of the Doppler spectrum at pulse repetition frequency (PRF) interval [18], a small number of "ghosts" often appear around in coastal areas, which can easily lead to a false detection, as exhibited in Fig. 1(b). The cross-polarization (HV and VH) channels tend to be less affected and used to remove azimuth ambiguities for ship detection purposes in some studies [19], [20]. In addition, SAR imaging relies on pulse compression, which results in strong scattering targets with obvious 2-D sidelobes. The sidelobe characters of clustered and isolated strong scattering points are shown in Fig. 1(c) and (d). And they appear more significant in the copolarization channel. The above indicates that single-polarization data has advantages and disadvantages under complex observation conditions. Only extracting object features based on single-polarization data cannot fully distinguish objects from the background, and the detection ability is still limited. Therefore, it is necessary to take advantage of the advantageous complementarity of dual-polarization data to extract fusion polarization characteristics and use them for a more complete object representation to improve the object detection ability in complex scenes.

Nevertheless, most CNN-based SAR object detection methods are designed for single-polarization data and cannot be directly employed for multipolarization data. This is mainly because the existing methods are mostly equipped with a single-pathway structure, which cannot simultaneously extract the features of the dual-polarization data and perform subsequent fusion operations on the dual-polarization features. Additionally, the existing common feature fusion algorithms [21], [22] mainly include elementwise summation, concatenation, elementwise product, weighted average, and bilinear product, which usually hold a weak fusion generalization. The simple application of these methods may be unable to filter out interference, lead to the inability to obtain robust results, even increase the burden on the network and reduce the discriminative ability of features. Therefore, it is necessary to design a novel network with multipathway structure to mine and integrate dual-polarization information for ship detection.

Aimed to mine, enhance, and fuse the polarization feature adaptively, a novel dual-polarization image feature fusion network (DPFF-Net) is proposed, of which the key modules include polarization-aware feature enhancement block (PAEB) and dynamic gated fusion block (DGFB). Copolarization and cross-polarization images serve as the network inputs, from which the deep hierarchical features are extracted through a siamese shared backbone of DPFF-Net. The shared backbone jointly mines and leverages the polarization features, enriching the structural integrity and building the high-level representations of ships. To further utilize and activate the extracted features, the PAEB is constructed to generate polarization-specific masks, which highlight the joint features sensitive to polarization attributes and strengthen the coupling relationship hidden in dual-polarization features. Considering that copolarization and cross-polarization images have various advantages in ship detection in complex scenes, DGFB is presented to fuse these two features further and adaptively balance their contributions. The design of DGFB, inspired by Zhang et al. [23] and Liu et al. [24], complements each other's advantages, effectively eliminating information differences caused by poor image quality.

Experiments on the Gaofen-3 dual-polarization ship detection dataset illustrate the effectiveness of the proposed DPFF-Net. Its focus is on the complementary information in different polarization images and balancing the two data's

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high-level features to establish robust feature representations for the detection head. It can eliminate some false positives (FPs) and effectively improve the overall detection performance.

The major contributions of this article are generalized in the following three aspects.

1) Different from the existing single-input methods, DPFF-Net is proposed to realize SAR ship detection. With copolarization and cross-polarization images as inputs, DPFF-Net extracts, enhances, and adaptively fuses deep hierarchical dual-polarization features through a siamese structure. It can effectively eliminate false alarms and improve the accuracy and recall of detection tasks.
2) To highlight the joint features sensitive to polarization attributes and strengthen the coupling relationship hidden in dual-polarization features, PAEB is devised to generate polarization-aware attention masks, guiding the activation and cross-enhancement (CE) of the feature maps from the siamese pathway.
3) To adaptively fuse and achieve the optimal enhanced features from two pathways, DGFB is designed to dynamically balance the various advantages of dual-polarization information, which complement each other's advantages, and effectively eliminate information differences caused by the poor image quality.

The remains of this article are arranged as follows. Section II introduces the related corresponding works briefly. Next, the proposed DPFF-Net is illustrated in Section III. Section IV describes the details of experiments and some analyses. Finally, the conclusions are presented in Section V.

II. Related Works

A. General Object Detection Based on Deep Learning

At present, common object detection methods are widely implemented by deep learning. This is mainly because deep convolutional networks in deep learning can learn robust and high-level feature representations of data. The current mainstream object detection methods can be divided into two-stage and one-stage approaches. The first stage of the two-stage approach begins by generating region proposals that contain rough location information for objects. These regions are further classified and refined in the second stage. The classic Faster R-CNN [25] abandons traditional candidate box generation methods, such as the selective search, sets different scale anchors, employs region proposal networks (RPNs) to extract candidate regions, and realizes end-to-end model training. Object detection via region-based fully convolutional networks (R-FCN) [26] cancels the region of interest (RoI)-wise subnetwork based on Faster R-CNN. It introduces ROIPooling and a position-sensitive score map to improve detection efficiency while retaining more spatial domain position information. Mask R-CNN [27] proposes RoIAlign, which uses the bilinear difference to replace the quantization operation of ROIPooling, effectively improving detection performance.

Compared with the two-stage methods, one-stage methods cancel the region proposal and directly regress objects' location coordinates and classification scores through one stage. YOLO [28] divides the input into grids of the same size and simultaneously predicts the bounding box and category of each grid. Its speed is also qualitatively improved. Subsequently, Joseph et al. [29] continue to improve on the basis of YOLO. YOLOv3, v5 [30], X [31] and other versions have been presented one after another. It should be noted that, as an anchor-free and nms-free detector, YOLOX has certain advantages in terms of performance and speed facing SAR data, so it is chosen as a basis for improvement in this article. Single shot multibox detector (SSD) [32] introduces the idea of multiscale features. Objects of different scales are predicted by features of different scales, significantly improving the detection accuracy of one-stage detectors. RetinaNet [33] introduces a practical "focal loss" to solve the problem of imbalance between positive and negative samples during training. This enables accuracy comparable to two-stage detectors while maintaining high detection speed. Methods such as CenterNet [34], CornerNet [35], fully convulsion one-stage object detector (FCOS) [36], Reppoints [37], etc., realize target detection by predicting some key points, avoiding designing anchors.

In addition, the transformer framework has been gradually introduced into object detection in the past two years. Combining CNN and transformer [38], Carion et al. [39] apply transformer to target detection and first propose a complete detection transformer (DETR). Zhu et al. [40] propose the deformable DETR model combined with deformable convolution.

B. CNN-Based SAR Object Detection Methods

Currently, CNN-based detection methods have shown superior performance in the field of SAR object detection [41]. At first, CNNs are employed for sea–land segmentation, candidate region extraction, and target identification in detection. Now most CNN-based detection algorithms use end-to-end mode, without preprocessing and sea–land segmentation. To better apply the CNN-based detection method to SAR object detection, Kang et al. [42] propose a multilayer fusion convolutional neural network detection algorithm based on context information, enhancing the detection performance of small-scale objects. Jiao et al. [43] densely connect features based on the Faster R-CNN framework to crack the issue of SAR ship detection in multiscale target distribution scenarios.

To further suppress false alarms and capture target features, some researchers adopt algorithms of attention mechanisms. Chen et al. [44] propose a detection model incorporated with an attention mechanism for accurate target localization in complex SAR scenes. Dense attention pyramid network (DAPN) [45] adopts the attention module with a pyramid structure in the feature extraction process, which is an effective method to detect multiscale ships in SAR images. Feature balancing and refinement network (FBR-Net) [46] proposes a feature balancing pyramid. It combines hierarchical and spatial attention mechanisms to balance features at different levels of semantic information. Cui et al. [47] designs a spatial transformation enhanced attention block. This block can extract more robust semantic features while suppressing the false alarms in some nearshore and inland disturbances.

In addition, starting from the SAR imaging mechanism, some researchers combine the scattering and polarization characteristics to realize some methods more suitable for SAR

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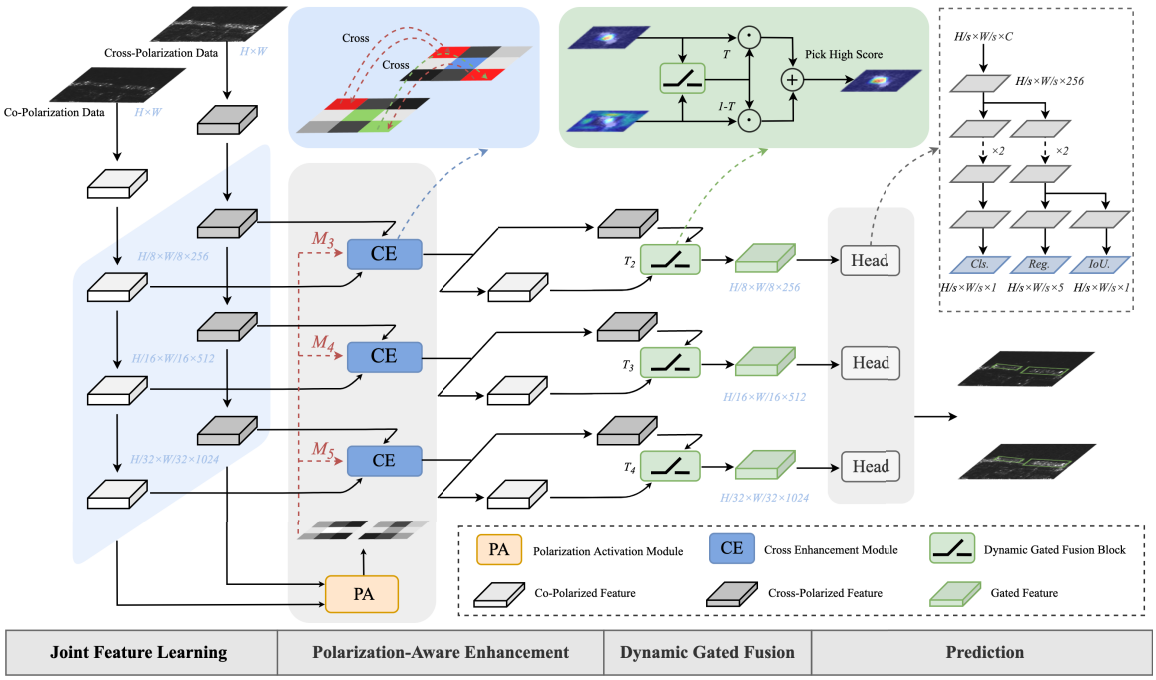

Fig. 2. Framework of the proposed DPFF-Net. It comprises a siamese shared backbone used for joint feature learning, PAEB, DGFB, and the prediction head.

target detection. Fu et al. [8] propose a scattering keypoint guided feature adaption method and locate objects with the oriented bounding box (OBB). By defining a set of scattering key points, the key areas belonging to targets can be depicted to locate the target more accurately. Xu et al. [48] introduce dual-polarization data, which generates three-channel pseudo-color images for polarization feature extraction and ship detection. Gao and Liu [49] employ compact polarimetric SAR data and the power–entropy decomposition to obtain polarization features fed into a RetinaNet-based network for predictions of ship targets.

The aforementioned methods can be summarized as follows:

$$R = P(E(S)) \tag{1}$$

where S represents an input SAR images, R is the detection result of the ships, $E(\cdot)$ and $P(\cdot)$ denote the feature extraction with multiple convolutional blocks and the prediction including regression, classification, etc., respectively. Although the above methods are suitable for most scenarios, there is still room for improvement in practical complex scenes, especially under poor image quality conditions. This is partly attributed to the fact that they mainly rely on single-polarization images for feature mining and object detection but lack the utilization of multipolarization information. Therefore, they are difficult to distinguish targets with similar amplitude features but significant differences in polarization features. In order to solve this problem, it is urgent to design a novel detection network, which is able to mine and fuse the dual-polarization information adaptively.

III. Proposed Method

To improve the upper bound of ship detection capability and suppress irrelevant false alarms in SAR images, the DPFF-Net is proposed. Its core lies in the joint extraction of differentiated polarization features, adaptive complementing and fusion of polarization information to obtain more comprehensive and accurate features for the detection head input, thus improving the overall performance. The overall framework of DPFF-Net is illustrated in Fig. 2. It comprises four parts: a siamese shared backbone used for joint feature learning, a PAEB, a DGFB, and a prediction head. More details are to be expounded in Sections III-A–III-D.

A. Overall Architecture

Compared with single-polarization images, dual-polarization images can supply more comprehensive polarization scattering information about targets, including many details such as target dielectric properties, geometry, and spatial orientation [50]. Reasonable mining and utilization of the above information will help highlight targets better, distinguish easily confused noise, and promote the upper bound of existing SAR ship detection algorithms with a single-pathway structure. Therefore, the DPFF-Net incorporating the siamese structure for SAR ship target detection is constructed. Its implementation form can be summarized as follows:

$$R = P(U(C(J(S^{\mathrm{co}}, S^{\mathrm{cr}})))) \tag{2}$$

where S^{co}, $S^{\mathrm{cr}} \in \mathbb{R}^{W\times H\times 3}$ indicate two kinds of SAR images at copolarization and cross-polarization with width W and height H. The operator $J(\cdot,\cdot)$, $C(\cdot)$, and $U(\cdot)$ denote the joint feature learning process, the polarization-aware feature enhancement, and dynamic gated fusion process, respectively.

Different from (1), the integrated siamese structure can process two types of data and extract two types of features simultaneously. The two parallel feature extraction

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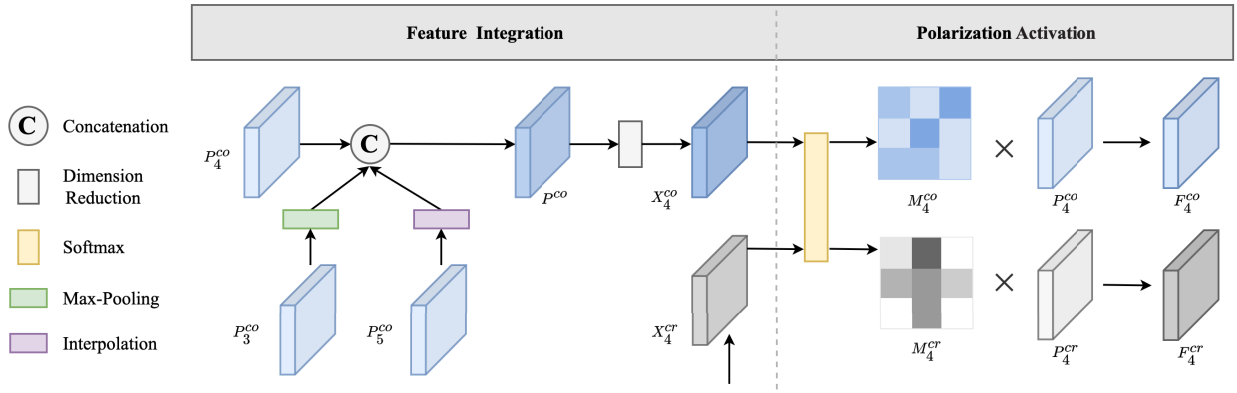

Fig. 3. Structure of the PA module. The feature maps at different stages in the copolarization and cross-polarization paths are integrated by concatenation after scaling to generate the merged features. These features are then operated in pairs by softmax operation to generate corresponding polarization-specific masks.

pathways are named the copolarization pathway and the cross-polarization pathway, respectively. Unlike the general siamese network that focuses on comparing the similarity between samples, the proposed DPFF-Net pays more attention to joint feature extraction and feature fusion to realize the adaptive optimization of detection features. Moreover, due to the application of PAEB and DGFB, the designed siamese structure is different from the existing siamese structure for object detection. It can better enhance the feature representation ability, adaptively select the feature information more conducive to fusion detection, and significantly improve the detection performance. PAEB and DGFB are detailed in the follow-up.

In addition, the proposed method is built on YOLOX pipeline and CSPDarknet53 [51] is used as the backbone. The detection head employs the structure of the decoupled head, which contains a 1×1 convolutional layer for feature dimensionality reduction and two independent parallel branches stacked by 3×3 convolutional layers for classification and regression. Among them, a intersection over union (IOU) branch is additionally embedded in the regression branch. Furthermore, we add angle regression to the regression branch, using PolyIoU loss for supervision to obtain rotated detection boxes.

B. Polarization-Aware Enhancement Block

As illustrated in the introduction, according to the SAR polarization characteristics, the same targets present different characteristics at two different polarizations. Therefore the PAEB is designed, which contains two submodules: polarization activation (PA) and CE. On the one hand, it spotlights features sensitive to polarization properties by generating polarization masks. On the other hand, it enables the intrinsic interaction of the features of two different polarizations in a way of cross attention.

1) Polarization Activation Module: In order to make the features sensitive to polarization properties more prominent and facilitate the representation ability of the network, the PA module is designed. It consists of feature integration and mask generation. According to the polarization attributes, it calculates the polarization-specific attention masks, which are utilized to guide activations of two kinds of features obtained by the siamese shared backbone.

Suppose $P_k \in \mathbb{R}^{W/s_k \times H/s_k \times C_k}$ are the feature maps at different stage k with $k = 3, 4, 5$, where C_k is the channel dimension of P_k, s_k is the corresponding downsampling rate to original input images, and $s_k = 2^k$. In the copolarization and cross-polarization pathway, the output features of the last three stages of feature extraction are represented as $\{P_3^{\mathrm{co}}, P_4^{\mathrm{co}}, P_5^{\mathrm{co}}\}$ and $\{P_3^{\mathrm{cr}}, P_4^{\mathrm{cr}}, P_5^{\mathrm{cr}}\}$. As shown in Fig. 3, we choose P_4^{co} as an intermediate feature and resize the remaining features in the same pathway to the same size with the interpolation and max-pooling strategies. These rescaled features are integrated by concatenation to generate merged features P^{co}. P^{co} is then fed into a 1×1 convolutional layer for its channel dimension reduction after going through max pooling and mean pooling, respectively. Here, generated $X_4^{\mathrm{co}} \in \mathbb{R}^{W/s_4 \times H/s_4 \times 1}$ can be formulated as

$$\begin{aligned} X_4^{\mathrm{co}} &= \mathcal{F}\left(\left[f^{\mathrm{Max}}(P^{\mathrm{co}}), f^{\mathrm{Mean}}(P^{\mathrm{co}})\right]\right) \\ &= f^{\mathrm{ReLU}}(f^{\mathrm{BN}}(f^{\mathrm{Conv}}(\left[f^{\mathrm{Max}}(P^{\mathrm{co}}), f^{\mathrm{Mean}}(P^{\mathrm{co}})\right]))) \end{aligned} \quad (3)$$

where $\mathcal{F}(\cdot)$ represents the 1×1 convolutional layer. $[\cdot, \cdot]$ means channel stitching, f^{Max} and f^{Mean} represent the max and mean pooling process, respectively. $f^{\mathrm{BN}}(\cdot)$ and $f^{\mathrm{ReLU}}(\cdot)$ denote the batch normalization operation and activation function, respectively. The cross-polarization pathway performs the same operation to generate the features X_4^{cr}. Then the integrated feature maps X_4^{co} and X_4^{cr} are spliced together and performed softmax operation [52] in pairs to get the corresponding polarization-specific masks $M_4^{\mathrm{co}} \in \mathbb{R}^{W/s_4 \times H/s_4 \times 1}$ and $M_4^{\mathrm{cr}} \in \mathbb{R}^{W/s_4 \times H/s_4 \times 1}$.

The obtained masks are then rescaled to the corresponding size using the same but reverse interpolation and max-pooling to gain the corresponding masks for the level 3 and 5. Finally, these polarization masks are multiplied with the corresponding points of the original feature maps separately. The output-activated features have the same size and dimension as the corresponding input features.

2) Cross Enhancement Module: After the copolarization and cross-polarization pathway features are activated by the PA module, to enhance the second-order correlation between the PA module's two kinds of activated features and learn the interactive hiding relationship between them, the CE module

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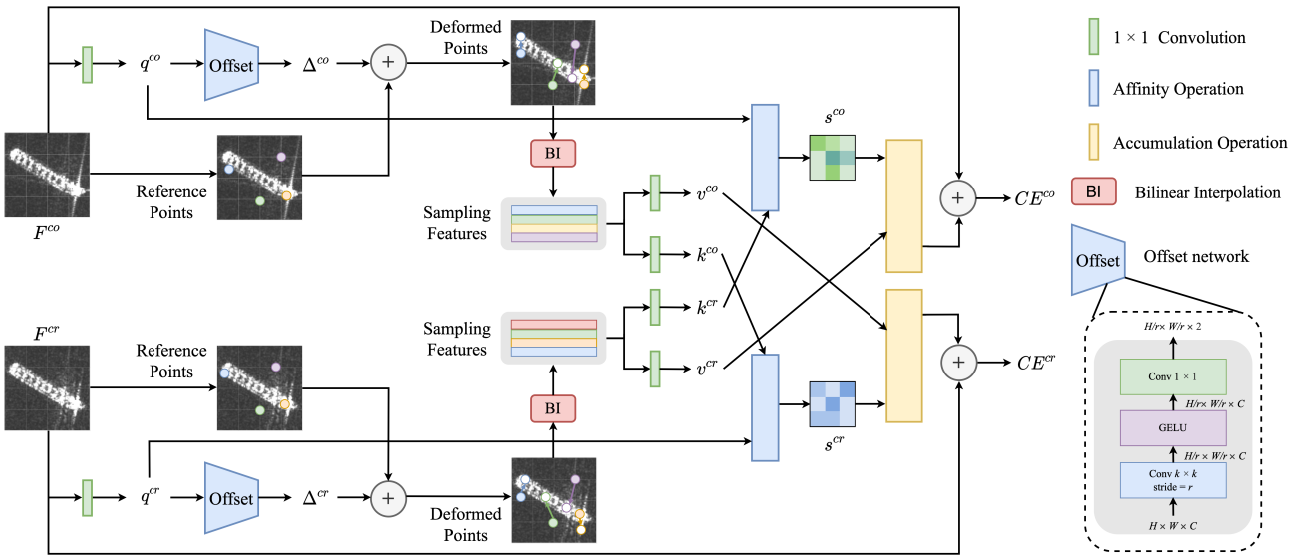

Fig. 4. Structure of the CE module. Two sets of reference points are, respectively, down-sampled from the feature maps in the two pathways. The feature maps in the two branches are transformed into queries by projection and then sent to the offset networks to learn the offsets of the reference points. The sampled features are obtained according to the deformed reference points, and their corresponding deformed keys and values are projected. Finally, the cross-enhanced features are generated through mutual calculations of the queries, deformed keys, and values. Only four reference points are placed in the figure for clarity. There are many more reference points in actual implementation.

is designed inspired by Xia et al. [53]. It is mainly divided into two parts: nonuniform feature sampling and cross attention.

Nonuniform feature sampling is achieved through offset learning and bilinear interpolation. As displayed in Fig. 4, given two activated feature maps F^{co}, $F^{\mathrm{cr}} \in \mathbb{R}^{w\times h\times c}$ are from the PA module, by downsampling from the two feature maps to generate two uniform grids of points p^{co}, $p^{\mathrm{cr}} \in \mathbb{R}^{w_r\times h_r\times 2}$ as references, in which r is the downsampling factor, $w_r = w/r$, $h_r = w/r$. These reference points are a series of linear coordinates $\{(0,0),\ldots,(h_r-1, w_r-1)\}$, and are normalized to the range $[-1,+1]$. Meanwhile, to get the offsets Δ^{co} and Δ^{cr} for each reference point, F^{co} and F^{cr} are linearly projected into q^{co}, q^{cr} via two 1×1 convolutions. Then q^{co}, q^{cr} go through the offset network $\theta_{\mathrm{offset}}(\cdot)$ to learn reasonable 2-D offsets

$$\Delta^{\mathrm{co}} = \theta_{\mathrm{offset}}(q^{\mathrm{co}}) \tag{4}$$

$$\Delta^{\mathrm{cr}} = \theta_{\mathrm{offset}}(q^{\mathrm{cr}}) \tag{5}$$

where Δ^{co}, $\Delta^{\mathrm{cr}} \in \mathbb{R}^2$ are of 2-D real vectors. Fig. 4 shows that two convolutional layers and a Gaussian error linear unit (GELU) activation constitute the offset network.

Then, the learned offsets are used for obtaining the nonuniform sampling positions

$$\hat{p^{\mathrm{co}}} = p^{\mathrm{co}} + \Delta^{\mathrm{co}} \tag{6}$$

$$\hat{p^{\mathrm{cr}}} = p^{\mathrm{cr}} + \Delta^{\mathrm{cr}}. \tag{7}$$

As $\hat{p^{\mathrm{co}}}$ and $\hat{p^{\mathrm{cr}}}$ are typically fractional, a sampling function $\Phi(\cdot;\cdot)$ via bilinear interpolation [54] $G(\cdot)$ is set to generate sampling features $\hat{F^{\mathrm{co}}} = \Phi\left(F^{\mathrm{co}}; \hat{p^{\mathrm{co}}}\right)$, $\hat{F^{\mathrm{cr}}} = \Phi\left(F^{\mathrm{cr}}; \hat{p^{\mathrm{cr}}}\right) \in \mathbb{R}^{w_r h_r\times c}$ according to the locations of deformed points

$$\begin{aligned}\Phi\left(F^{\mathrm{co}}; (p_x, p_y)\right) &= \sum_{(l_x,l_y)} G\left((l_x,l_y),(p_x,p_y)\right)\cdot F^{\mathrm{co}}[l_x,l_y] \\ &= \sum_{(l_x,l_y)} g(l_x,p_x)g\left(l_y,p_y\right)\cdot F^{\mathrm{co}}[l_x,l_y] \\ &= \sum_{(l_x,l_y)} \max(0,1-|l_x-p_x|) \\ &\quad\times \max\left(0,1-\left|l_y-p_y\right|\right)\cdot F^{\mathrm{co}}[l_x,l_y]\end{aligned} \tag{8}$$

where $(p_x, p_y) \in \hat{p^{\mathrm{co}}}$, (l_x, l_y) enumerates all integral spatial locations closest to (p_x, p_y) in the feature map F^{co}, $g(m,n) = \max(0, 1-|m-n|)$ and $G(\cdot)$ is nonzero only for these locations. $F^{\mathrm{co}}[l_x, l_y]$ represents the value of the location (l_x, l_y) in the feature map F^{co}. The sampling operation for F^{co}, it is not described in this article. Then the feature $\hat{F^{\mathrm{co}}}$ are converted into keys ($k^{\mathrm{co}} \in \mathbb{R}^{w_r h_r\times c}$) and values ($v^{\mathrm{co}} \in \mathbb{R}^{w_r h_r\times c}$) by linear projection. The cross-polarization pathway performs the same operation to generate the k^{cr} and values v^{cr}.

With q^{co}, q^{cr}, k^{co}, v^{cr}, k^{cr}, and v^{cr}, the similarity matrix s^{co} is computed by the query q^{co} associated with the copolarization and the key k^{cr} associated with cross-polarization. Meanwhile, the similarity matrix s^{cr} is computed by the query q^{cr} associated with cross-polarization and the key k^{co} associated with copolarization. Finally, the cross enhanced features $\mathrm{CE}^{\mathrm{co}}$ in the copolarization pathway assisted by the cross-polarization is weighted by s^{co}, while the cross enhanced features $\mathrm{CE}^{\mathrm{cr}}$ in the copolarization pathway is weighted by s^{cr}. The calculations involved above can be summarized as

$$\begin{aligned}\mathrm{CE}^{\mathrm{co}} &= \gamma s^{\mathrm{co}} v^{\mathrm{cr}} + F^{\mathrm{co}} \\ &= \gamma\,\mathrm{Softmax}\left(q^{\mathrm{co}}(k^{\mathrm{cr}})^T\right)v^{\mathrm{cr}} + F^{\mathrm{co}}\end{aligned} \tag{9}$$

$$\begin{aligned}\mathrm{CE}^{\mathrm{cr}} &= \gamma s^{\mathrm{cr}} v^{\mathrm{co}} + F^{\mathrm{cr}} \\ &= \gamma\,\mathrm{Softmax}\left(q^{\mathrm{cr}}(k^{\mathrm{co}})^T\right)v^{\mathrm{co}} + F^{\mathrm{cr}}\end{aligned} \tag{10}$$

where γ is a scalar parameter, it is set as 0.75 by default in the experiments. Through the cross attention between two polarization features, the sampled content feature and corresponding reference element from two different pathways can interact better with each other and have a finer-grained feature representation.

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C. Dynamic Gated Fusion Block

Through the above PAEB, to explore the complementary and commonality of the enhanced features from two pathways and make them better adaptive fusion, we specially design the dynamic gated fusion module (DGFB) to combine copolarization and cross-polarization features effectively for detection.

To be specific, the DGFB is divided into three layers, including a concatenation operation, a convolution layer, and a sigmoid layer. Given two activated feature maps $\mathrm{CE}^{\mathrm{co}}$, $\mathrm{CE}^{\mathrm{cr}} \in \mathbb{R}^{w\times h\times c}$ are from PAEB, respectively. After concatenation, a fused feature map $P^{\mathrm{fu}} \in \mathbb{R}^{w\times h\times 2c}$ can be gained. Then the convolution layer with weights $w \in \mathbb{R}^{c\times 1\times 1\times 2c}$ (c filters with a dimension of $1 \times 1 \times 2c$ per filter) is used to learn the correlations of the two feature maps and weigh their contributions for the subsequent prediction. The output of the convolution layer is a coefficient matrix $T \in \mathbb{R}^{w\times h\times c}$ with the value

$$T_{x,y,n} = \sum_{m=1}^{2C} P^{\mathrm{fu}}_{x,y,m} \times w_{n,x,y,m} \quad n \in [1,c], \quad x \in [1,w], \quad y \in [1,h]. \tag{11}$$

Then the sigmoid layer is applied to regularize T to keep $T_{x,y,n} \in [0,1]$. $T^{\mathrm{co}} = T$ and $T^{\mathrm{cr}} = 1 - T$ are termed the weighted gates, which represent the respective decisiveness of the copolarization and cross-polarization features for predicting the target. The two weighted gates are utilized to weigh the contributions of copolarization and cross-polarization features as

$$\hat{\mathrm{CE}}^{\mathrm{co}} = \mathrm{CE}^{\mathrm{co}} * T^{\mathrm{co}} \tag{12}$$

$$\hat{\mathrm{CE}}^{\mathrm{cr}} = \mathrm{CE}^{\mathrm{cr}} * T^{\mathrm{cr}} \tag{13}$$

where * represents Hadamard product. Finally, we generate the gated fusion feature map as

$$\hat{P}^{\mathrm{fu}} = \hat{P}^{\mathrm{co}} + \hat{P}^{\mathrm{cr}}. \tag{14}$$

We predict the object by the fused features.

D. Loss Function

The training loss function L of the designed model consists of object loss L_{obj}, classification loss L_{cls}, and regression loss L_{obbox}, which are defined as follows:

$$L = L_{\mathrm{obj}} + L_{\mathrm{cls}} + L_{\mathrm{obbox}}. \tag{15}$$

Among them, the object and classification loss use the cross-entropy loss function [55]

$$L(p_i, \hat{p}_i) = -[p_i \log(\hat{p}_i) + (1 - p_i)\log(1 - \hat{p}_i)] \tag{16}$$

where p_i and $\hat{p}_i$ represent ground truth and the model's predicted value, respectively. After obtaining decoded rotated bounding boxes, the regression loss adopts PolyIoU loss in [28], defined as follows:

$$L_{\mathrm{obbox}} = 1 - \mathrm{PolyIoU} \tag{17}$$

where PolyIoU represents the intersection ratio between predicted orientation values and label values.

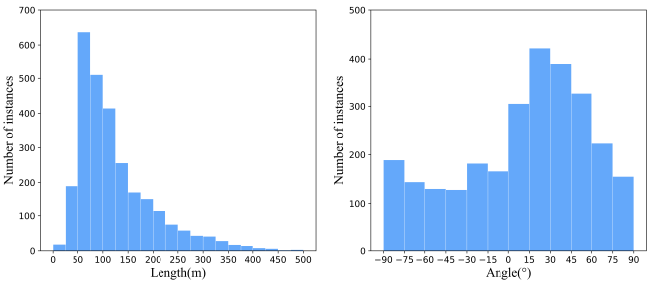

Fig. 5. Distribution of scale and direction of the labeled instances.

IV. Experiments

This section revolves around experiments conducted on 20 SAR images (ten pairs). Section IV-A presents the specifics related to the experimental data. The implementation details of the experiments are given in Section IV-B, and the evaluation metrics are described in Section IV-C. The ablation experiments and validity analyses of the proposed siamese shared backbone (denoted as JFL-Subnet), PAEB, and DGFB are reported in Section IV-D. Section IV-E delivers further analyses of the proposed method. The proposed method is compared with the state-of-the-art methods in Section IV-F.

A. Experimental Data Description

20 SAR images (ten pairs) with a spatial resolution of 2.5 m are employed as experimental data to investigate the effectiveness of DPFF-Net. Each pair consists of pictures in HV and HH polarization modes. Among them, 14 images (seven pairs) are employed for training and validation, and the remaining six images (three pairs) are appointed as test data. These data are provided by the C-band SAR remote sensing satellite Gaofen-3. The incidence angle ranged from 23.87° to 50.42°. The sea-state condition is relatively acceptable. These data cover four ports (multitemporal), including Shanghai, Tianjin, Zhoushan, and Tokyo. Their size ranges from 25 607 × 27 491 pixels to 26 507 × 28 308 pixels. The experimental dataset contains 2759 multiscale and multiangle labeled instances, whose statistical distribution is depicted in Fig. 5. All instances in these data have been professionally labeled and inspected, and these can be used as a basis for evaluating the subsequent experiments.

B. Implementation Details

The designed DPFF-Net is implemented using the PyTorch framework [39]. Each raw large SAR image is cropped into patches with three different sizes (384 × 384, 512 × 512, 600 × 600) in a sliding window mode to enhance the generalization ability. At the same time, each patch is guaranteed to contain at least one target. In the end, approximately 7960 patches (3980 per polarization) are generated. During model training, the size of these patches is uniformly adjusted to 384 × 384. The model is trained for 60 epochs on an 8-GB NVIDIA RTX2080 Ti GPU. The batch size and initial learning rate per image are 8 and 0.00125, respectively. During inference, a sliding window with a stride of 96 is used to generate 384 × 384 patches from the original image and feed them to the model for detection. The predictions of

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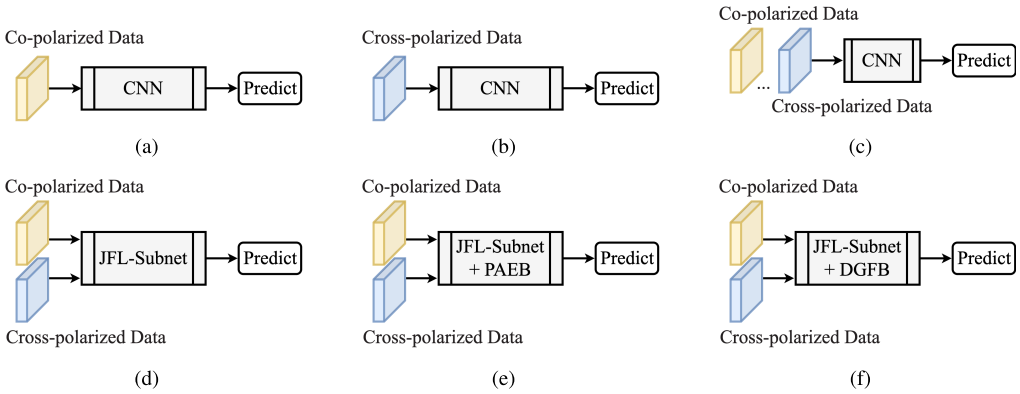

Fig. 6. Schematic diagrams of the ablation experiment. (a) Baseline A. (b) Baseline B. (c) Baseline C. (d) Baseline+. (e) Baseline+ PAEB. (f) Baseline+ DGFB.

each patch are combined to get the final detection result of the original image. Additionally, some overlapping predicted rotated boxes are filtered by the non-maximum suppression (NMS) algorithm with an IoU threshold of 0.1.

C. Evaluation Metrics

Standard metrics such as precision, recall, F1-Measure, and average precision (AP) are employed to evaluate the model performance reasonably. The calculation of these metrics is related to the IoU, which represents the degree of overlap of the predicted results with the ground truths. If the IoU is above a set threshold, the corresponding predicted box is considered a true positive (TP). On the contrary, if the IoU is lower than the set threshold, it is recorded as an FP. If a labeled box has no matching predicted box, it is marked as a false negative (FN). After counting the numbers C_{TP}, C_{FP}, and C_{FN} of TP, FP, and FN, respectively, the calculation of precision and recall can be deduced

$$\text{precision} = \frac{C_{\mathrm{TP}}}{C_{\mathrm{TP}} + C_{\mathrm{FP}}} \tag{18}$$

$$\text{recall} = \frac{C_{\mathrm{TP}}}{C_{\mathrm{TP}} + C_{\mathrm{FN}}}. \tag{19}$$

Precision is the proportion of TPs in all predicted results, and recall is the proportion of TPs in the ground truths. The evaluation index F1-measure [56] is jointly calculated by precision and recall to reconcile the two

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{20}$$

Different recall and precision can be obtained according to different confidence thresholds. With recall as the abscissa and precision as the ordinate, a 2-D curve can be drawn, named the precision recall (PR) curve. The area under the PR curve is the AP measure, often used to evaluate the quality of the detection model comprehensively, and is calculated as follows:

$$\mathrm{AP} = \int_0^1 P(R)\, dR. \tag{21}$$

AP value is positively correlated with model performance. The AP metric used in this article is the AP value at IoU = 0.5, which is recorded as $\mathrm{AP}_{0.5}$.

D. Ablation Studies

In this section, a series of ablation experiments are performed using the dataset mentioned in Section IV-A to analyze the effectiveness of the proposed method and each component. Their schematic diagrams are exhibited in Fig. 6(a)–(f). All ablation experiments share the same setup and training strategies. The specific experimental settings are as follows.

1) Baseline A, B, and C: Baseline A, B, and C all adopt a single-pathway structure. The data sources of Baseline A and B are HH and HV polarization amplitude data, respectively. The data source of Baseline C is achieved by a simple mixture of the amplitude data of HH and HV polarization without distinguishing their polarization modes and performing any superposition or concatenation operations on them. Its data volume is twice that of Baseline A and B.

2) Baseline+: Baseline+ is equipped with JFL-Subnet, a dual-pathway feature extraction structure. These two pathways are named copolarization pathway and cross-polarization pathway, respectively. In Baseline+, HH and HV polarization amplitude data are paired into the network by sharing the backbone to extract their respective in-depth features simultaneously.

3) Baseline+ PAEB: This model is embedded in a series of PAEBs in the copolarization and cross-polarization pathway, and the other settings are identical to the Baseline+.

4) Baseline+ DGFB: A sequence of DGFBs are added in the neck of the model, the rest of the settings are the same as Baseline+.

5) DPFF-Net: DPFF-Net holds a siamese structure to jointly capture the features at two different polarizations. Additionally, in this model, a sequence of PAEBs are cascaded for mutually reinforcing the two types of data features, and the DGFBs for the gated fusion of two kinds of features.

The results of all ablation studies above are recorded in Table I. The evaluation metric in the last column of Table I represents inference time on a single cropped patch.

6) Effect of JFL-Subnet: It can be known from Table I that the F1 and AP scores of Baseline+ model with JFL-Subnet are 0.6639 and 0.5515, respectively. Compared with Baseline A, which takes HH polarization images as inputs, the Baseline+ increases the F1 and AP by 1.30% and 3.59%, respectively. Compared with the Baseline B, which takes

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TABLE I
INFLUENCE OF EACH COMPONENT IN THE PROPOSED METHOD

Model	Input	PAEB	DGFB	Recall	Precision	F1	AP	Time(ms)
Baseline A	HH			0.7013	0.6151	0.6554	0.5321	23.4
Baseline B	HV			0.7088	0.6305	0.6674	0.5406	23.1
Baseline C	HH & HV			0.6852	0.6232	0.6527	0.5293	23.3
Baseline +	HH & HV			0.7152	0.6194	0.6639	0.5515	25.3
Baseline + PAEB	HH & HV	✓		0.7498	0.6887	0.7180	0.6109	27.9
Baseline + DGFB	HH & HV		✓	0.7388	0.6761	0.7061	0.5993	26.5
DPFF-Net (Ours)	HH & HV	✓	✓	**0.7684**	**0.7428**	**0.7554**	**0.6437**	28.3

TABLE II
ABLATION STUDIES OF PAEB

PA	CE	Recall	Precision	F1	AP
		0.7152	0.6194	0.6639	0.5515
✓		0.7227	0.6542	0.6867	0.5663
	✓	0.7291	0.6672	0.6968	0.5905
✓	✓	**0.7498**	**0.6887**	**0.7180**	**0.6109**

TABLE III
RESULTS OF DIFFERENT FUSION FORMS

Network	Recall	Precision	F1	AP
Baseline + CWCF	0.7152	0.6194	0.6639	0.5515
Baseline + EWSF	0.7211	0.6004	0.6552	0.5376
Baseline + EWPF	0.7053	0.6217	0.6631	0.5433
Baseline + DGFB	**0.7388**	**0.6761**	**0.7061**	**0.5993**

HV polarization images as inputs, the Baseline+ increases the AP by 2.02%, while keeping F1 comparable. The above comparison experiments show that the joint use of copolarization and cross-polarization data helps improve the model performance, which is attributed to the complementary information of the two types of data and the increase of the total information. Relying on JFL-Subnet to reasonably extract two types of data feature information is the key to improving the detection task, and it has achieved apparent advantages in performance. Since the Baseline+ model uses both polarization images, the amount of data is increased. Under the premise of ensuring that the amount of data is equivalent, to verify the role of JFL-Subnet in the feature extraction stage, the experimental comparison between Baseline C and Baseline+ is carried out. Currently, the data is not fed into the network in pairs as in Baseline+ but is trained in an unordered state. The results show that Baseline+ is better than Baseline C 1.72% and 4.14% in F1 and AP, respectively. Comparing Baseline C with Baseline A or B shows that simply mixing dual-polarization images to increase the amount of data and using single-pathway structure is not stronger than using single-polarization data, and even weakens the performance. This result verifies the feasibility of leveraging the copolarization and cross-polarization images to build an SAR ship detection pipeline. Also, it reflects the rationality of using JFL-Subnet with a siamese structure in the proposed method.

7) Effect of PAEB: Table I shows that the PAEB increases the AP and F1 by 8.15% and 10.77%, respectively, compared with the Baseline+. To investigate the effectiveness of the two submodules in PAEB, comparative experiments are conducted by adding PA and CE module individually to the Baseline+. The experimental results are presented in Table II. PA module increases the AP and F1 by 2.68% and 3.43%, respectively. CE module increases the AP and F1 by 7.07% and 4.97%, respectively. The results indicate that PA and CE modules have a positive impact on the improvement of model performance, and the joint use of these two modules can further contribute to the final impact. Copolarization and cross-polarization images can provide unique semantic information. PA module can make the characterization of different polarization-aware importance more prominent so that the complementarity can be better exploited later. Meanwhile, the CE module can learn mutual information from the two branches, making the two branches work more collaboratively.

8) Effect of DGFB: Table I shows the gain of DGFB for detection performance. To analyze the effectiveness of DGFB, some additional feature fusion strategies of the DPFF-Net backend are formulated, and comparative experiments are carried out on this basis. The feature fusion forms used include channelwise concatenation fusion in Baseline+, simple elementwise summation fusion, and elementwise product fusion. They are named as Baseline+ CWCF (the same as Baseline+), Baseline+ EWSF, and Baseline+ EWPF, respectively. Their schematic diagrams are shown in Fig. 7(a)–(c), and their parameter settings are duplicates of DGFB. The ablation studies of DGFB are reported in Table III. From Table III, DGFB reaches the highest scores on all evaluation metrics compared to the rest of the fusion strategies. Meanwhile, it improves 6.36%–7.77% F1, 8.67%–11.48% AP over Baseline+ CWCF, Baseline+ EWSF, and Baseline+ EWPF, respectively. These results indicate that oversimplified fusion forms have weak fusion ability and do little to help fuse in-depth features of different polarizations. In contrast, DGFB has a more adaptive form of feature reorganization and is conducive to obtaining more robust detection performance.

E. Discussion

In this part, we will have some further discussions about the proposed DPFF-Net. By some additional comparison experiments and the qualitative analyses, the effectiveness of three main contributions in our method is shown in different scenes.

1) Analysis of the Joint Feature Learning: The joint feature learning serves as the feature extraction strategies used for

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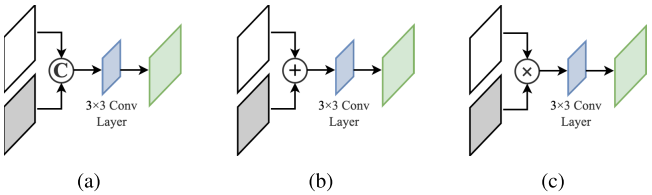

Fig. 7. Some different feature fusion forms for DGFB validation. The white and gray diamond boxes represent copolarization and cross-polarization features, respectively. (a) Baseline+ CWCF: channelwise concatenation fusion dimension. (b) Baseline+ EWSF: elementwise summation fusion. (c) Baseline+ EWPF: elementwise product fusion.

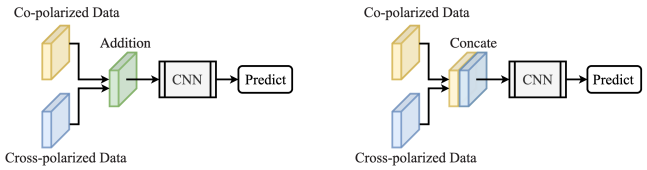

Fig. 8. Some strategies of feature extraction of copolarization and cross-polarization data. (a) Strategy 1: elementwise addition. (b) Strategy 2: concatenation in channel dimension.

TABLE IV
ANALYSIS OF SOME FEATURE EXTRACTION STRATEGIES ON THE VALIDATION SET

Strategy	Input	Recall	Precision	F1	AP
Strategy 1	HH & HV	0.6641	0.6263	0.6447	0.5087
Strategy 2	HH & HV	0.6929	0.6418	0.6664	0.5355
Baseline C	HH & HV	0.6852	0.6232	0.6527	0.5293
Baseline +	HH & HV	**0.7152**	**0.6194**	**0.6639**	**0.5515**

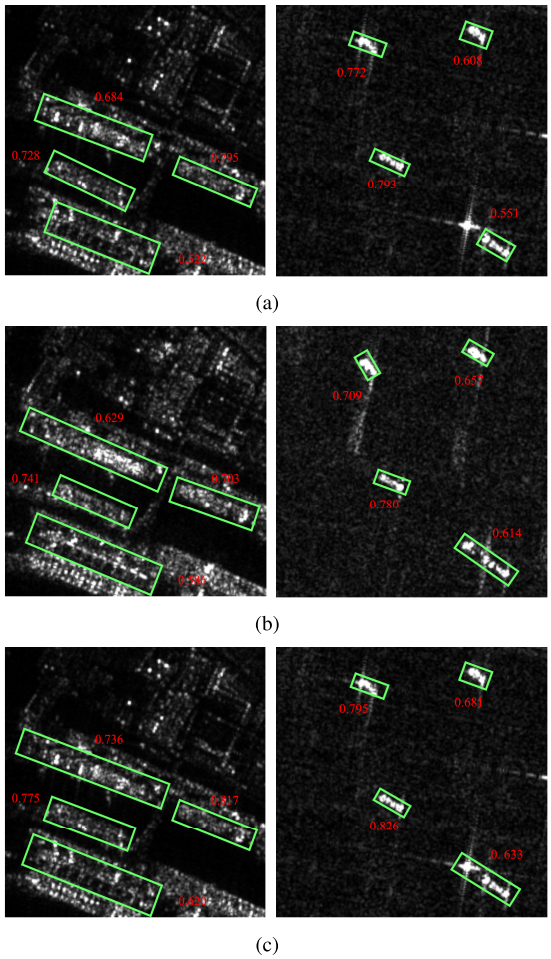

Fig. 9. Different detection results of networks with and without PAEB. The green rectangles visualize the predicted boxes of the model. The red numbers in these images describe the confidence of the target. (a) Results of Baseline A. (b) Results of Baseline B. (c) Results of Baseline+ PAEB.

dual-polarization data in the proposed method. Various feature extraction strategies of copolarization and cross-polarization data may determine the ability of feature representations for SAR ship detection. To investigate the proper strategy for feature extraction structure, some experiments are carried out on other feature extraction structures. Here, we consider two early-fusion methods to deal with the two kinds of data: the addition of pixel values and the concatenation in channel dimension. Fig. 8 illustrate these structures of two strategies: Strategy 1 and Strategy 2. Other settings of these models are the same. Table IV exhibits the experimental result. Strategy 1 shows the performance of 0.6447 F1 and 0.5087 AP, whose performance is degradative compared with Baseline A and B with a single-pathway structure. In contrast, Baseline+ improves F1 and AP by 2.98% and 8.41%, respectively. This result verifies that the direct addition of pixels does nothing to improve detection ability. When the two kinds of SAR image data are directly superimposed, the target will become more significant, but the noise will also be more awful, and the interference such as azimuth ambiguities and speckle noise cannot be filtered out. The Strategy 2 shows the performance of 0.6664 F1, 0.5355 AP. Its performance is slightly improved compared to Baseline A and B, but it is still not as good as Baseline+. The reason is that the concatenation in the channel dimension drives the information carried by the two polarization data intermixed at the initial stage. It is demanding for this intermixed input to approximate the supervisory signal. Therefore, it is challenging for the model to grasp the complementary relationship between pairs of SAR images, and the training process is prone to fall into local optimum.

For SAR images, copolarization and cross-polarization data can provide more complementary and discriminative information for targets and their surroundings. However, these early fusion methods fail to effectively use the complementary relationship between the two kinds of data and may even lead to negative optimization. The experiment results also confirmed that the simple, early fusion method (Strategy 1 and 2) and unordered input (Baseline C) have little benefit in improving the model's performance when the amount of input information is comparable. It also verifies that the JFL-Subnet with a fusion at the feature level effectively constructs more sufficient feature representations of targets for SAR ship detection.

2) Analysis of PAEB: To illustrate the impact of PAEB more intuitively, some visual detection results are generated and presented in Fig. 9. The first and second rows in Fig. 9 are the detection results of Baseline A and Baseline B, respectively. In other words, they are the detection results of the model

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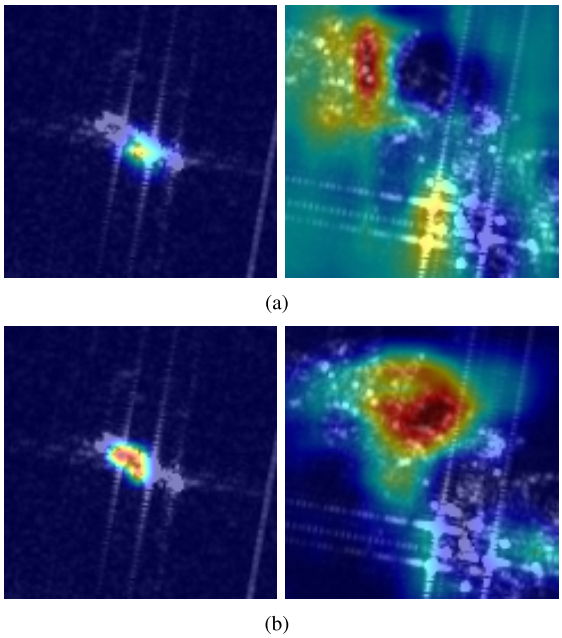

Fig. 10. Visual analysis of the features extracted from various models. (a) Model without DGFB. (b) Model with DGFB.

trained on copolarization and cross-polarization data, respectively. It can be observed that some ships in both polarization images have lower confidence in the interference of complex near-shore background or imaging ambiguity. It is challenging for detectors to concentrate on accurate ships. PAEB focuses on features sensitive to polarization characteristics by generating polarization masks and makes copolarization and cross-polarization features interact internally through cross-attention, thus enhancing the target features. After introducing PAEB, the detection results are loaded on the copolarization images, as pictured in Fig. 9(c). It can be noticed that the application of PAEB makes the capture of the contour structure of targets more complete, the targets' location more accurate, and the targets' confidence significantly improved.

3) Analysis of DGFB: To analyze the impact of DGFB more intuitively, Fig. 10 displays the visual feature maps. Here, GradCAM [57] is used to visually explain the output part of the DGFB. Grad-CAM utilizes the gradients streaming into the final convolution layer of networks to highlight the crucial parts inputs for inferencing results. As shown in Fig. 10(a), ships on the sea surface have lower activation scores, and certain parts of the port that are easily confused as targets still have higher activation scores. Only through simple mid-term fusion strategies the in-depth features of different polarizations cannot be well integrated to focus on real objects, and the feature discrimination is not strong enough. By embedding DGFB in the polarization feature fusion at the network's backend, the disturbed targets docked in the port in Fig. 10(b) receive more attention and evolve more discriminative. Interference around them can be largely suppressed. There are two reasons for the advancement obtained by DGFB. On the one hand, it combines elementwise addition strategy, puts more emphasis on feature commonality, which can capture the correlation and diversity between features simultaneously. On the other hand, the polarization features are balanced and fused in a dynamic fashion. The copolarization and cross-polarization information in the two corresponding paths can be effectively utilized to eliminate the information differences and redundant features, so that the objects are more discriminative against distractors and background.

TABLE V
COMPARISON WITH VARIOUS CNN-BASED DETECTION METHODS

Method	Recall	Precision	F1	AP
RRPN $_{co}$	0.6040	0.5218	0.5599	0.5295
RRPN $_{cross}$	0.6342	0.5439	0.5856	0.5496
RRPN	0.6273	0.5210	0.5692	0.5273
RetinaNet OBB $_{co}$	0.6176	0.5061	0.5563	0.5168
RetinaNet OBB $_{cross}$	0.6142	0.5287	0.5683	0.5470
RetinaNet OBB	0.5899	0.5116	0.5533	0.5374
FCOS OBB $_{co}$	0.6076	0.5219	0.5555	0.5322
FCOS OBB $_{cross}$	0.6281	0.5389	0.5801	0.5618
FCOS OBB	0.6102	0.5295	0.5670	0.5403
YOLOv5 OBB $_{co}$	0.6098	0.5519	0.5794	0.5575
YOLOv5 OBB $_{cross}$	0.6453	0.5676	0.6040	0.5827
YOLOv5 OBB	0.6193	0.5658	0.5913	0.5674
Proposed DPFF-Net	**0.7684**	**0.7428**	**0.7554**	**0.6437**

F. Comparison With State-of-the-Art Methods

To validate the superiority of DPFF-Net, we compare it with other state-of-the-art CNN-based methods and conduct a sequence of comparative experiments. Since these advanced methods process SAR images employing a single pathway and have yet to fully consider the issues mentioned in Section I, experiments are conducted by taking the copolarization and cross-polarization information data as input separately. At the same time, to ensure the consistency of the total amount of input information, the two polarization data are only mixed disorderly for training as the Baseline C.

The experimental results are reported in Table V. The subscripts co and cr represent the use of copolarization and cross-polarization data, respectively. No subscript means that the two kinds of data are mixed. These CNN-based methods in Table V employ copolarization, cross-polarization, and an unordered mixture of these two types of data. They may retain limited performance and generalization ability for ship detection in dual-polarization data. Compared to these methods using copolarization data for training and inference, DPFF-Net outperforms them by 25.07%–36.53% F1 and 10.47%–24.55% AP. Furthermore, DPFF-Net shows a relatively significant improvement of 7.04%–12.74% F1 and 7.04%–12.74% AP compared to these methods using cross-polarization data. These results verify the significance of the rational utilization of multipolarization information in DPFF-Net.

Fig. 11 shows the comparison of the prediction results of DPFF-Net and YOLOv5 OBB. It is not difficult to notice that the proposed DPFF-Net has better detection performance and relatively strong anti-interference ability in complex scenes. In contrast, in the inshore environment with complex background interference and dense distribution of ships, the number of false alarms is reduced, and the detection rate is improved after using DPFF-Net. In the offshore scenario, the proposed method maintains adequate detection performance in a simple background. At the same time, it has a more vital ability

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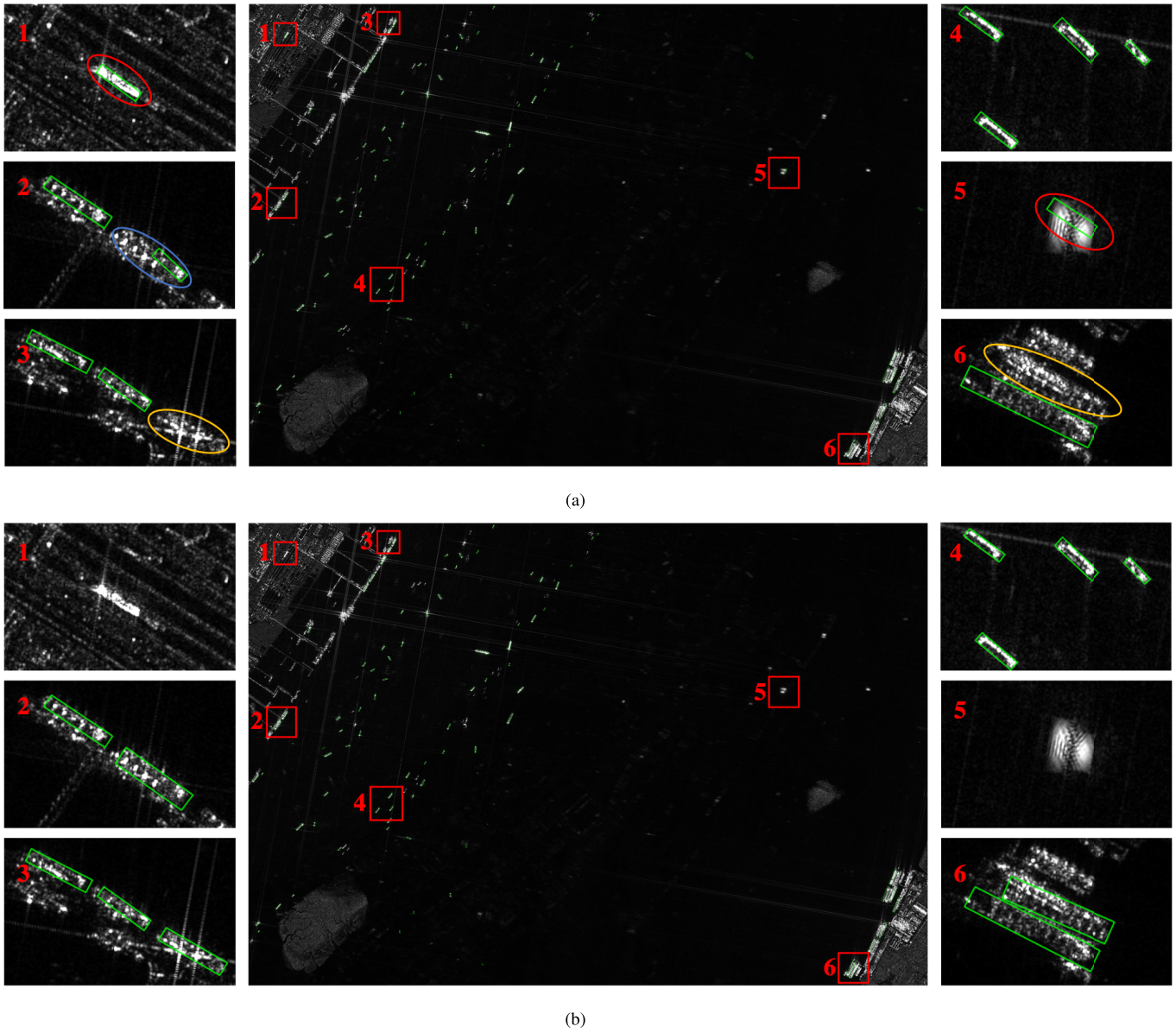

Fig. 11. Comparison of the prediction results of the proposed DPFF-Net and YOLOv5 OBB. The green boxes display prediction results. The red, yellow, and blue oval boxes represent false targets, undetected ships, and ships with inaccurate localization. (a) YOLOv5 OBB. (b) Proposed DPFF-Net.

to suppress false alarms when facing confusing false targets on the sea surface, such as the "ghosts" caused by azimuth ambiguities.

Despite the significant progress of our research, there are still some limitations. The polarization characteristics of small ships (generally less than 40 m) are relatively fuzzy, and the differences between different polarization modes are not noticeable. It is easy to lose effective information in the feature extraction stage, and the detector is challenging to capture the complementary characteristics between the features. As a result, this method is unsuitable for ships with too small size. It also inspires our future work.

V. Conclusion

A novel SAR ship detection method, DPFF-Net, is proposed to promote the upper bound of SAR ship detection performance in complex sea scenes. With specific designed siamese structure, DPFF-Net can effectively mine the correlation and complementary information hidden in dual-polarization images and achieve more comprehensive and accurate features used as the detect head input. PAEB and DGFB are two core modules devised to realize the abovementioned function. The former highlights the joint features sensitive to polarization attributes and strengthens their coupling relationship. The latter is proposed to fuse dual-polarization features and balance their contributions adaptively. The whole design complements the advantages of various polarization information and improves the overall performance. The ablation experiments validate the reasonability and effectiveness of the proposed modules. And the comparison with other splendid detectors on Gaofen-3 datasets verifies the superiority of our proposed method.

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